Atakan Topaloğlu

+41 77 290 22 17 | atakantopaloglu@gmail.com | Homepage | GitHub | Zürich/Switzerland

Education

ETH Zürich | Zürich *September 2025 – Expected Graduation: June 2027*
MSc. in Information Technology and Electrical Engineering, GPA: 5.59/6.00
Selected Coursework: Probabilistic AI, Inference and Estimation
IEEE Signal Processing Society Scholarship (renewed)

Koç University | Istanbul *September 2021 – June 2025*
BSc. in Electrical and Electronics Engineering, GPA: 3.99/4.00, Department Rank: 2/135
Selected Coursework: Digital Logic Design, C Programming, Signals & Systems, DSP
IEEE Signal Processing Society Scholarship **(first ever recipient from a Turkish University)**

Work & Research Experience

Flexion Robotics | Incoming AI Engineering Intern (Series A, $50M Funding) *May 2026 – Oct 2026*
• Selected for a highly competitive internship to engineer and deploy C++ reinforcement learning policies onto humanoid hardware. Will focus on real-time execution constraints and hardware-software integration.

ETH Zürich | Research Assistant, Photogrammetry & Remote Sensing Lab *September 2025 – Present*
• Recruited as Research Assistant at MSc start; redesigned point-cloud preprocessing by replacing repeated per-instance NumPy queries with a hash-based point-to-object index, reducing runtime from **8 hours to <2 minutes (~250× speedup).**
• Eliminated I/O bottleneck (evidenced by <1% GPU/CPU utilization) by moving data augmentation offline via multiprocessing and HDF5, and accelerated radius-based point filtering with axis-aligned bounding squares to reject distant points early, cutting end-to-end training from **12 days to 8 hours (~36× speedup).**

Google | Research Collaborator *March 2025 – September 2025*
• Set a **new state-of-the-art on sparse-view Gaussian Splatting**. We grounded generative priors with an uncertainty oracle to filter generative artifacts. Implemented the custom CUDA kernel to enable depth supervision in **3DGS MCMC.**
• Initiated and led the project, from proposal to first author WACV 2026 Oral acceptance (top 3% of submissions).

Technical University of Munich | Undergraduate Summer Researcher *July 2024 – September 2024*
• Engineered a modified neural audio codec for vibrotactile data compression, achieving >30x compression ratios while maintaining signal fidelity (PSNR). **Optimized the decoder for real-time CPU execution**.
• Proposed and designed experiments for a novel spatial masking approach to enhance the Multi-Channel Vibrotactile Codec, which turned into an article, awaiting acceptance (*Supervisor: Prof. Eckehard Steinbach*; Erasmus+ Internship).

Siemens | Part-Time R&D Working Student *August 2023 – July 2025*
• Designed a 3D reconstruction pipeline for 3DGS, **reducing peak GPU VRAM usage by 83%** to enable processing on consumer hardware (Nvidia Quadro P1000, 4GB VRAM) via heavy color palette quantization and subsampling post SfM.
• Implemented a viewer that runs efficiently without a GPU, by refactoring data layout from **AoS to SoA** to **maximize cache hits and enable SIMD**, and implementing **octrees + frustum culling** to eliminate non-visible geometry.
• Developed a wavelet-based SVM classifier to identify defects in A-Scan ultrasound data for non-destructive wind turbine testing. Secured a ~$20k industrial ultrasound kit for trial at no cost, resolving a 4-month sourcing delay.

Skills

Programming Languages and Frameworks: Python, PyTorch, C++17/20, CUDA, Nsight, ONNX, Docker, PostgreSQL
Languages (Spoken): German (Intermediate), English (TOEFL: 119/120), Turkish (Native)
Interests: Chess (ELO: 1823), Formula 1&Telemetry, Running

Selected Projects & Publications

• OracleGS (WACV 2026, Oral): First author. Solved inverse problems in 3D reconstruction using generative priors and uncertainty estimation. Code released in PyTorch/CUDA.
• Super-Resolution Evaluation (ICIPW 2025): First author. Proposed a difficulty-aware metric for evaluating signal reconstruction models, demonstrating standard metrics (PSNR) fail to capture edge-case performance.